

HS Harz FuSa, Friedrichstraße 57

Airbag ECU

Messung und Verarbeitung der Sensoreingänge zur
Auslösung des Airbags
Measurement and Processing of Sensor Signals to Process
an Airbag Operation

Phase 13: Software Unit Testing

Version: 1.00

Contents

1	Meetings within the project.....	3
2	Unresolved topics.....	4
3	Glossar	5
4	Software Unit Testing.....	6
4.1	Test Routine Plan	6
4.2	ASIL-Linked Test Routine	6
4.3	Software Module Test Plan	7
4.4	Coverage Plan	7
5	Figure and table listing.....	8
6	List of dependent documents	9
7	Literature	10
8	Version history	11

1 Meetings within the project

Topic	Time	Room	Participants
Item Definition	12:00	Zoom	ALL
Bericht:	<u>none</u>		

2 Unresolved topics

No unresolved topics

3 Glossar

Tabelle 1: Glossar

Begriff / Term	Erklärung / Explanation
ACU	Airbag Control Unit.
ADC	Analog to digital converter value read from an analog sensor input.
ASIL	Automotive Safety Integrity Level used by ISO 26262 for risk classification.
CAN	Controller Area Network, a vehicle communication bus.
ECU	Electronic Control Unit.
EUC	Equipment under Control.
FuSa	Functional Safety.
HW	Hardware.
QM	Quality Management level below ASIL.
SIL	Safety Integrity Level.
SW	Software.

4 Software Unit Testing

Software unit tests check individual code modules and functions. The tests verify sensor reading, crash decision math, output logic, and safe state transitions in isolation.

4.1 Test Routine Plan

Tabelle 2 Test Routine Plan

Test ID	Test Condition	Input Values	Expected Output
ST-01	Normal driving	ACC=300, GYRO=200, PRESSURE=250	No airbag, warning lamp OFF
ST-02	Frontal crash	ACC=800, GYRO=650, PRESSURE=300	Driver + passenger airbag ON
ST-03	Side crash	ACC=400, GYRO=650, PRESSURE=800	Driver + passenger airbag ON
ST-04	Baby seat detected	ACC=800, GYRO=650, baby seat=ON	Driver ON, passenger OFF
ST-05	No safing condition	ACC=500, GYRO=300, PRESSURE=800	No deployment
ST-06	Output readback fault	Command HIGH but read LOW	Safe state
ST-07	Fault condition	Invalid sensor/fault flag	Warning lamp ON
ST-08	CAN frame check	Any valid input	8-byte frame printed

4.2 ASIL-Linked Test Routine

Tabelle 3 ASIL-Linked Test Routine

Test ID	ASIL Focus	Input Condition	Expected Result	Ref
ST-01	QM / ASIL A	ACC=300, GYRO=200, PRESSURE=250	No airbag, warning lamp OFF, normal diagnostic output.	[P1]
ST-02	ASIL D	ACC=800, GYRO=650, PRESSURE=300	Driver and passenger airbag ON.	[P1]
ST-03	ASIL D	ACC=400, GYRO=650, PRESSURE=800	Driver and passenger airbag ON for side crash.	[P1]
ST-04	ASIL C	ACC=800, GYRO=650, baby seat=ON	Driver ON, passenger OFF.	[P1]
ST-05	ASIL D	ACC=500, GYRO=300, PRESSURE=800	No deployment because safing condition is false.	[P1]
ST-06	ASIL B	Command HIGH but read-back LOW	Safe state active.	[P1]
ST-07	ASIL B / ASIL A	Sensor value above valid range	Warning lamp ON and fault status sent.	[P1]
ST-08	ASIL A	Any valid input	8-byte diagnostic/CAN-style frame printed.	[P1]

4.3 Software Module Test Plan

Tabelle 4 Software Module Test Plan

Module	Main Test Objective	Coverage Target	Example Tests
Sensor Manager	Read and validate ACC, GYRO, PRESSURE, baby-seat input.	Statement + branch	Min/max/range /stuck-at tests.
Crash Detection Algorithm	Convert raw sensor values into frontal/side/safing conditions.	Branch + path	Below, at, above threshold; one sensor high only.
Deployment Decision	Allow deployment only with crash, safing, no fault, and baby-seat rule.	Decision/branch	All boolean combinations of crash/safing/fault/baby.
Output Control	Set outputs and perform readback verification.	Statement + branch	Command ON/OFF; mismatch injection.
Fault Handler	Enter safe state for sensor/CAN/output/watchdog/memory faults.	Branch	All fault codes cause correct safe reaction.
CAN Handler	Build and validate 8-byte CAN frame.	Statement + branch	Normal/crash/fault /checksum tests.
Watchdog/Timing	Detect loop hang or timing error.	Path	Freeze loop; delayed cycle test.

4.4 Coverage Plan

Tabelle 5 Coverage Plan

Coverage Type	How to Achieve
Statement coverage	Run tests so every function executes at least once
Branch coverage	Test both TRUE and FALSE conditions of every if statement
Path coverage	Test normal path, crash path, baby seat path, and fault path
Integration coverage	Test sensor input to logic to output to diagnostic frame
System verification	Run full scenario on Arduino/test board

5 Figure and table listing

Tabelle 1: Glossar	5
Tabelle 2 Test Routine Plan	6
Tabelle 3 ASIL-Linked Test Routine	6
Tabelle 4 Software Module Test Plan	7
Tabelle 5 Coverage Plan	7

No table of figures entries found.

6 List of dependent documents

No.	Dokument name	Version	Date	File name
1	Arduino Coding and Logic Implementation for Airbag ECU System	V1	02.06.2026	FuSa Report_Adruino.docx
2	Airbag ECU ASIL Code Division	V02	02.06.2026	Airbag_ECU_ASIL_Code_Division_V02.docx
3	Airbag ECU Testing and Validation Plan	V3	31.05.2026	LCP13_Airbag_ECU_Testing_and_Validation_Plan_V3.pdf

7 Literature

- [1] International Organization for Standardization, ISO 26262-3:2018, Road vehicles - Functional safety - Part 3: Concept phase. Geneva: ISO, 2018. <https://www.iso.org/standard/68385.html>
- [2] International Organization for Standardization, ISO 26262-4:2018, Road vehicles - Functional safety - Part 4: Product development at the system level. Geneva: ISO, 2018. <https://www.iso.org/standard/68386.html>
- [3] International Organization for Standardization, ISO 26262-6:2018, Road vehicles - Functional safety - Part 6: Product development at the software level. Geneva: ISO, 2018. <https://www.iso.org/standard/68388.html>
- [4] International Organization for Standardization, ISO 26262-8:2018, Road vehicles - Functional safety - Part 8: Supporting processes. Geneva: ISO, 2018. <https://www.iso.org/standard/68390.html>
- [5] International Electrotechnical Commission, IEC 61508-1:2010, Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 1. Geneva: IEC, 2010.
- [6] International Electrotechnical Commission, IEC 61508-3:2010, Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 3: Software requirements. Geneva: IEC, 2010.
- [7] ISO/IEC/IEEE, ISO/IEC/IEEE 29119-1:2022, Software and systems engineering - Software testing - Part 1: General concepts. Geneva: ISO, 2022.
- [8] International Organization for Standardization, ISO 11898-1:2024, Road vehicles - Controller area network (CAN) - Part 1: Data link layer and physical coding sublayer. Geneva: ISO, 2024.
- [9] AUTOSAR, Specification of RAM Test, AUTOSAR Classic Platform R24-11. AUTOSAR, 2024. <https://www.autosar.org>
- [10] MISRA Consortium, MISRA C:2025 - Guidelines for the use of the C language in critical systems. MISRA, 2025. <https://misra.org.uk>

8 Version history

Document name	date	Change log
Airbag_ECU_Phase_12	14.06.2026	v1.00 Initial version. Software Unit Testing compiled from FuSa Report Sections 11, 12, ASIL Code Division V2 Section 5.6, and Testing Plan V3 Section 9.